

12.3 Dot Product

1 Warm Up

Problem 1: If $a = \langle 2, -5 \rangle$, find $|a|$.

Problem 2: Find $\cos(\pi/4)$, $\cos(\pi/2)$, $\cos(\pi/3)$.

2 Motivation

Now that we have unpacked all the definitions corresponding to vectors, we can be creative with them! Wouldn't it be cool if we had two vectors a and b and we defined the "product" of a and b . This is in fact novel since we only know how to multiply a vector by a scalar. What would $a \cdot b$ mean and what would be the most logical way to multiply two vectors?

3 Definitions & Theorems

1. If $a = \langle x_1, y_1, z_1 \rangle$ and $b = \langle x_2, y_2, z_2 \rangle$, We say the **dot product** $a \cdot b = x_1x_2 + y_1y_2 + z_1z_2$.
2. Question, What is $a \cdot a$? Answer: $|a|^2$.
3. $a \cdot (b + c) = a \cdot b + a \cdot c$
4. Okay this seems pretty cool, but does the dot product mean anything. Yes, geometrically $a \cdot b$ is a scaled measure of the angle between your vectors. More precisely we can prove that $a \cdot b = |a||b|\cos(\theta)$ where θ is the angle between a and b between 0 and π . This can be proved using the Law of Cosines.
5. Perpendicular vectors are also called **orthogonal** and two vectors are orthogonal if and only if $a \cdot b = 0$.
6. Suppose we have a vector $a = \langle a_1, a_2, a_3 \rangle$. If we scale or "normalize" a vector by its magnitude we get $\frac{1}{|a|}a = \langle a_1/|a|, a_2/|a|, a_3/|a| \rangle$ which is a vector of magnitude 1 or a **unit vector**. All unit vectors lie on our unit circle. We also use our dot product theorem and replace b with $i = \langle 1, 0, 0 \rangle$ and see that $\cos(\alpha) = a_1/|a|$. Thus, our normalized vector gives us some sense of direction called the **direction cosines** or **direction angles** of a which are $a/|a| = \langle \cos(\alpha), \cos(\beta), \cos(\gamma) \rangle$.
7. The **vector projection** of b onto a is the vector with initial point at the initial point of a and terminal point on vector a where the perpendicular projection of the terminal point of b hits the vector a . We write $\text{proj}_a b$. The signed magnitude of

the vector projection is called the **scalar projection** of b onto a . This is denoted $\text{comp}_a b$. We have the following equations from simply looking at the cosine (A/H) of a right triangle.

Theorem 1. *The scalar projection of b onto a :*

$$\text{comp}_a b = \frac{a \cdot b}{|a|}.$$

Theorem 2. *The vector projection of b onto a :*

$$\text{proj}_a b = \frac{a \cdot b}{|a|^2} a$$

8. We can use the dot product to find work when you have a force vector F and a displacement vector D . We then have $\text{Work} = F \cdot D$.

4 Examples

Problem 1. *Find the dot product $\langle 2, -5, 1 \rangle \cdot \langle 2, 1, -4 \rangle$.*

Problem 2. *Find the dot product $u \cdot v$ where $u = 3i + 3j + 3k$ and $v = 2i - 2j$.*

Problem 3. *Find the angle between $\langle 1, 0, -4 \rangle$ and $\langle -2, -3, -1 \rangle$.*

Problem 4. Find the direction cosines of $\langle 3, -10, 2 \rangle$.

Problem 5. Find the direction angles of $u = 2i - 4j + k$.

Problem 6. Find the scalar projections and vector projections of $b = \langle 1, -3, -3 \rangle$ onto $a = \langle 0, -2, 5 \rangle$

Problem 7. I walk my dog up a 45 meter hill at a constant force of 28N. The leash is held at an angle of 22° . Find the work done pulling my dog up the hill.

5 Scratch Work